```
 1
 2 % write your own function for the DFS algorithm.
 3 function [retmap,retvisited,retsteps] = dfs(mapfile,startlocation,targetlocation)
 4
 5     % Convert the map file into a MATLAB matrix
 6     map = map_convert(mapfile);
 7     retmap = map;
 8
 9     % Get dimensions of the map
10     [maprows,mapcols] = size(map);
11
12     % Create a matrix to keep track of visited cells
13     visited = false(maprows,mapcols);
14
15     % Keep track of the parent of every visited cell.
16     parent = zeros(maprows,mapcols,2);
17
18     % DFS stack
19     stack = startlocation;
20
21     % Mark the starting location as visited
22     visited(startlocation(1),startlocation(2)) = true;
23
24     % Store the order in which cells were visited
25     retvisited = [];
26
27     % Used to indicate whether target has been found
28     found = false;
29
30     % Directions:
31     %       up
32     %       down
33     %       left
34     %       right
35     directions = [
36         -1  0
37          1  0
38          0 -1
39          0  1
40     ];
41
42     % DFS
43     while ~isempty(stack)
44
45         % Take the last item from the stack
46         current = stack(end,:);
47         stack(end,:) = [];
48
49         % Record that this cell was visited
50         retvisited = [retvisited; current];
51
52         % Have we reached the target?
53         if current(1) == targetlocation(1) && current(2) == targetlocation(2)
54             found = true;
55             break;
56         end
57
58         % Check each neighbouring cell
59         for k = 1:4
60
61             neighbour = current + directions(k,:);
```

```
62
63              row = neighbour(1);
64              col = neighbour(2);
65
66              % Check that neighbour is inside the map
67              if row >= 1 && row <= maprows && col >= 1 && col <= mapcols
68
69                  % Check that it is not a wall
70                  if map(row,col) == 0
71
72                      % Check that it hasn't already been visited
73                      if ~visited(row,col)
74
75                          % Mark as visited
76                          visited(row,col) = true;
77
78                          % Remember where we came from
79                          parent(row,col,1) = current(1);
80                          parent(row,col,2) = current(2);
81
82                          % Add neighbour to stack
83                          stack = [stack; neighbour];
84
85                      end
86                  end
87              end
88          end
89      end
90
91      % Reconstruct the path from target back to start
92
93      retsteps = [];
94
95      if found
96
97          current = targetlocation;
98
99          while ~(current(1) == startlocation(1) && current(2) == startlocation(2))
100
101             % Add current cell to the path
102             retsteps = [current; retsteps];
103
104             % Move to the parent
105             parentrow = parent(current(1),current(2),1);
106             parentcol = parent(current(1),current(2),2);
107
108             current = [parentrow parentcol];
109         end
110
111         % Add the starting location
112         retsteps = [startlocation; retsteps];
113
114     end
115
116     plotmap(retmap, retsteps);
117
118 end
119
120 function placestep(position,i)
121 % This function will plot a insert yellow rectangle and also print a number in this
    rectangle. Use with plotmap/viewmap.
```

```
position = [16-position(1) position(2)];

position=[position(2)+0.1 position(1)+0.1];

rectangle('Position',[position,0.8,0.8],'FaceColor','y');

c=sprintf('%d',i);

text(position(1)+0.2,position(2)+0.2,c,'FontSize',10);

end

```